



UMBC

Quantum Game Theory

Equilibrium Search, Optimization, and Learning

John Bui

1. Project Framing
2. Classical Background
3. Quantum Background
4. Quantum Game Theory
5. Related Paper Analysis
6. Research Direction
7. Learning Theory
8. Takeaways

Project Framing

We aim to analyze equilibria in classical and quantum strategic games by investigating these theoretical questions:

1. Nash equilibrium after introducing entanglement
2. Eisert-Wilkens-Lewenstein (EWL) protocol
3. Equilibrium search: learning or optimization problem?

Classical Background

Table 1: Possible scenarios from actions {Cooperate, Defect}.

Decision	Outcome
1. Both cooperate	Reward: both banned for 1 month
2. One defects, one cooperates	Temptation: defector receives warning; Sucker's payoff: cooperator permabanned
3. Both defect	Punishment: both banned for 1 year

Classical prisoner's dilemma (cont.)

Ban duration is inversely related to payoff. Define

$$T = 3, \quad R = 2, \quad P = 1, \quad S = 0,$$

so that

$$T > R > P > S.$$

Clyde		Clyde cooperates	Clyde defects
Bonnie			
Bonnie cooperates	$R, 2$	$R, 2$	$S, 0$
Bonnie defects	$T, 3$	$S, 0$	$P, 1$

Figure 1: Payoff matrix for Bonnie and Clyde.

Unlike prisoner's dilemma, a game of chicken contains multiple Nash equilibria and strategic coordination behavior.

$P_1 \backslash P_2$	P_2 swerves	P_2 straight
	P_1 swerves	P_1 straight
swerves	0, 0	-1, 1
straight	1, -1	-1000, -1000

Figure 2: Payoff matrix for the game of chicken played by players P_1 and P_2 .

A strategy profile s is a **Nash equilibrium** if no player can improve by changing strategy alone:

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}).$$

A strategy profile is **Pareto efficient** if no other outcome makes one player better off without making another worse off.

In prisoner's dilemma:

(D, D) is Nash equilibrium,

but

(C, C) is Pareto superior.

This paradox is known as the prisoner's dilemma.

Quantum Background

Quantum computing processes information using principles from quantum mechanics, especially:

- **Superposition**
- **Entanglement**
- **Measurement**

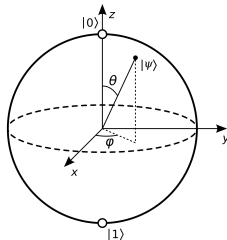
Goal

Quantum game theory studies how quantum phenomena alter strategic decision-making and equilibrium behavior.

0 ●

1 ●

Classical bit stores either 0 or 1



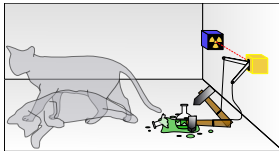
Bloch sphere of a qubit

A qubit may exist in a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

A quantum state may contain multiple possible outcomes simultaneously until measured, collapsing the state probabilistically:

$|0\rangle$ with probability $|\alpha|^2$, $|1\rangle$ with probability $|\beta|^2$.



Schrödinger's cat: superposition before observation

Interpretation

Quantum systems evolve continuously until measurement.

Two quantum systems are **entangled** when their joint state cannot be separated into independent components:

$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle.$$

This means:

- measuring one system reveals information about the other,
- correlations exist beyond classical independence,
- strategies may become correlated through entanglement.

Connection to Quantum Game Theory

Entanglement changes the strategic correlation structure of a game and may alter equilibrium behavior.

Quantum Game Theory

Classically, a player chooses:

C or D .

In the quantum version, the basis states represent classical actions:

$|C\rangle = \text{cooperate}, \quad |D\rangle = \text{defect}.$

A general quantum state may be a superposition:

$$|\psi\rangle = \alpha|C\rangle + \beta|D\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Players do not directly choose outcomes. They apply unitary operations to qubits, and outcomes appear only after measurement.

Eisert, Wilkens, and Lewenstein proposed the following protocol:

1. Start with the classical state:

$$|CC\rangle.$$

2. Apply an entangling operator J .
3. Each player applies a local unitary strategy:

$$U_A \otimes U_B.$$

4. Undo the entanglement with $J^\dagger$, then measure.

The final state is:

$$|\psi_f\rangle = J^\dagger(U_A \otimes U_B)J|CC\rangle.$$

The entangling operator is commonly written as:

$$J(\gamma) = \exp\left(-i\frac{\gamma}{2}D \otimes D\right), \quad \gamma \in \left[0, \frac{\pi}{2}\right].$$

$\gamma = 0 \Rightarrow$ classical game.

$\gamma = \frac{\pi}{2} \Rightarrow$ maximally entangled quantum game.

Interpretation

The parameter γ controls the transition from independent classical choices to correlated quantum strategic behavior.

Classical

(D, D)


 $\gamma = 0$

Transition

correlations emerge



Quantum

(Q, Q)


 $\gamma = \frac{\pi}{2}$

Interpretation

Increasing entanglement changes the strategic correlation structure of the game, shifting equilibrium behavior from classical defection toward cooperative quantum equilibria.

Players choose unitary transformations of the form:

$$U(\theta, \phi) = \begin{pmatrix} e^{i\phi} \cos(\theta/2) & \sin(\theta/2) \\ -\sin(\theta/2) & e^{-i\phi} \cos(\theta/2) \end{pmatrix}.$$

Classical cooperation and defection are special cases:

$$C = I, \quad D = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

EWL also identifies a quantum strategy:

$$Q = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

Players act on qubits in the joint Hilbert space:

$$\{|CC\rangle, |CD\rangle, |DC\rangle, |DD\rangle\}.$$

	$ C\rangle_{P_2}$	$ D\rangle_{P_2}$
$ C\rangle_{P_1}$	$ CC\rangle$	$ CD\rangle$
$ D\rangle_{P_1}$	$ DC\rangle$	$ DD\rangle$

Measurement collapses the final quantum state into one of these classical outcomes.

After measurement, the final quantum state produces probabilities for the four classical outcomes:

$$P_{CC}, \quad P_{CD}, \quad P_{DC}, \quad P_{DD}.$$

Player A's expected payoff is:

$$\mathbb{E}[u_A] = RP_{CC} + SP_{CD} + TP_{DC} + PP_{DD}.$$

Player B's expected payoff is:

$$\mathbb{E}[u_B] = RP_{CC} + TP_{CD} + SP_{DC} + PP_{DD}.$$

For fixed opponent strategy U_B , player A solves:

$$U_A^* \in \arg \max_{U_A \in \mathcal{S}} \mathbb{E}[u_A(U_A, U_B; \gamma)].$$

A Nash equilibrium is a pair (U_A^*, U_B^*) such that:

$$\mathbb{E}[u_A(U_A^*, U_B^*)] \geq \mathbb{E}[u_A(U_A, U_B^*)],$$

$$\mathbb{E}[u_B(U_A^*, U_B^*)] \geq \mathbb{E}[u_B(U_A^*, U_B)].$$

Learning-Theoretic View

The strategy space is continuous and matrix-valued, so equilibrium search resembles nonconvex policy optimization rather than finite action selection.

To make the theory testable, the project can simulate the EWL game over a grid of strategy parameters:

1. Sweep entanglement values $\gamma \in [0, \frac{\pi}{2}]$
2. Grid-search player strategies $U(\theta, \phi)$
3. Compute $|\psi_f\rangle = J^\dagger(U_A \otimes U_B)J|CC\rangle$
4. Convert amplitudes into outcome probabilities:
 $P_{CC}, P_{CD}, P_{DC}, P_{DD}$
5. Plot expected payoffs and approximate best responses

Expected Output

Increasing γ changes best-response behavior and when (Q, Q) becomes stable under the restricted EWL strategy set.

Related Paper Analysis

Main contribution: Meyer showed that a player with access to quantum strategies can outperform a classical opponent in a simple strategic game.

Relevance:

- One of the earliest examples of quantum advantage in game theory.
- Superposition can behave differently from classical randomization.

Relevance

Meyer's paper motivates the idea that strategy spaces can be expanded by quantum mechanics, and introduces the broader question: **what happens when all players have quantum strategies?**

Main contribution: EWL introduced a quantum version of prisoner's dilemma. Under *maximal* entanglement, the classical equilibrium (D, D) can be replaced by the quantum equilibrium (Q, Q) . At (Q, Q) , both players receive the cooperative payoff

$$(R, R).$$

Interpretation

Entanglement allows stable cooperation *without* communication.

Main contribution: Benjamin and Hayden argued that the EWL result depends strongly on the restricted set of allowed strategies. If players are allowed to use a larger set of unitary operations, then:

- the proposed quantum equilibrium may no longer be stable,
- counter-strategies may exist,
- the cooperative resolution becomes less automatic.

Relevance

This critique turns the project from “quantum games solve prisoner’s dilemma” into a more careful question about assumptions, strategy spaces, and equilibrium stability.

Research Direction

Parameterize each player's strategy by (θ, ϕ) and update parameters using payoff feedback:

$$(\theta_A, \phi_A) \leftarrow (\theta_A, \phi_A) + \eta \nabla_{(\theta_A, \phi_A)} \mathbb{E}[u_A].$$

$$(\theta_B, \phi_B) \leftarrow (\theta_B, \phi_B) + \eta \nabla_{(\theta_B, \phi_B)} \mathbb{E}[u_B].$$

Connection to Statistical Learning

This resembles multi-agent policy-gradient learning: agents iteratively update parameterized strategies using observed payoff signals.

Deliverables

Worst-case: rigorous theoretical comparison and literature review.

Best-case: payoffs and learning trajectories from equilibrium search.

Learning Theory

A player searching for a good quantum strategy is solving an optimization problem:

$$\operatorname{argmax}_{U_A} \mathbb{E}[u_A(U_A, U_B)].$$

In repeated play, agents could update strategies similar to:

- reinforcement learning,
- policy optimization,
- gradient-based learning,
- multi-agent learning.

Interpretation

Quantum equilibrium search can be viewed as learning over a continuous unitary strategy space.

A classical mixed strategy randomizes over actions:

$$pC + (1 - p)D.$$

A quantum strategy transforms amplitudes:

$$|\psi\rangle = \alpha|C\rangle + \beta|D\rangle.$$

The difference is that amplitudes can interfere before measurement.

Theoretical Claim

Unlike classical randomization, entanglement and interference can produce correlations unavailable to independent mixed strategies.

Takeaways

- **Restricted strategy spaces:** EWL's cooperative equilibrium depends on which unitary strategies are allowed.
- **Physical implementation:** Real quantum systems are affected by noise and decoherence.
- **Equilibrium stability:** Enlarging the strategy set may introduce counter-strategies.
- **Learning dynamics:** It is unclear whether adaptive agents would actually converge to the quantum equilibrium.
- **Interpretation:** Quantum advantage may depend on whether entanglement is treated as a resource provided before the game.

Open Problem

When does quantum game theory produce new strategic behavior rather than a reformulation of correlated/mixed classical strategies?

The EWL protocol, the first pioneer of quantum game theory, suggests that entanglement can convert the inefficient classical equilibrium of prisoner's dilemma into a cooperative quantum equilibrium.

However, the Benjamin-Hayden critique shows that this conclusion is not unconditional. The stability of the result depends on how the quantum strategy space is defined.

Key Takeaway

Quantum game theory serves as a theoretical bridge between equilibrium analysis, optimization, and learning.

Next steps:

- implement payoff-landscape plots over (θ, ϕ, γ) ,
- visualize whether increasing γ changes best-response behavior,
- test when (Q, Q) is stable under the restricted EWL strategy set.

Scope

Although a theoretical project, my goal is connecting the theory to statistical learning via computation/practical implementation.

- [1] David A. Meyer. *Quantum strategies*. Phys. Rev. Lett., 82(5):1052-1055, 1999.
- [2] Jens Eisert, Martin Wilkens, and Maciej Lewenstein. *Quantum games and quantum strategies*. Phys. Rev. Lett., 83(15):3077-3080, 1999.
- [3] Simon C. Benjamin and Patrick M. Hayden. *Comment on "Quantum games and quantum strategies"*. Phys. Rev. Lett., 87(6), 2001.
- [4] John Nash. *Non-cooperative games*. Annals of Mathematics, 54(2):286-295, 1951.
- [5] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018.
- [6] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.
- [7] Lucian Busoniu, Robert Babuska, and Bart De Schutter. *A comprehensive survey of multiagent reinforcement learning*. IEEE Trans. Syst., Man, Cybern. C, 38(2):156-172, 2008.
- [8] Richard Jankowski. *Punishment in iterated chicken and prisoner's dilemma games*. Rationality and Society, 2(4):449-470, 1990.
- [9] Sau-Him Paul Lau and Vai-Lam Mui. *Using turn taking to achieve intertemporal cooperation and symmetry in infinitely repeated 2×2 games*. Theory and Decision, 72(2):167-188, 2012.
- [10] Kazuki Ikeda and Shoto Aoki. *Infinitely repeated quantum games and strategic efficiency*. Quantum Information Processing, 20(12), 2021.
- [11] Thomas S. Ferguson. *A Course in Game Theory*. World Scientific, 2020.

Thank You!